Try It Yourself

Chapters 2–3 of Flutter and Dart Fundamentals (Dart B) · Original exercise wording

CHAPTER 2 / DART FOUNDATIONS

Try It Yourself · textbook page 26

01 · Enhanced Calculator

Modify the calculator program to also display the result of integer division (~/) and modulo (%). Make sure it handles the division by zero case for all three division operations.

02 · Temperature Converter

Write a program that reads a temperature in Celsius from the user and prints the equivalent in Fahrenheit. The formula is F = (C × 9/5) + 32. Use tryParse() to handle invalid input gracefully.

03 · Null Safety Practice

Write a program that asks the user for two inputs: their name and their age. Handle the case where the user provides no name (use ?? to provide a default of “Anonymous”) and the case where the age isn’t a valid number (use tryParse() and print an appropriate message). If both inputs are valid, print a greeting using string interpolation: “Hello, [name]! You will be [age + 1] next year.”

Classroom selection: Core tasks are Temperature Converter and Null Safety Practice. Enhanced Calculator is an extension. These labels are instructor guidance, not part of the textbook exercise wording.

Try It Yourself

Chapter 3 · Control Flow and Loops

CHAPTER 3 / CONTROL FLOW AND LOOPS

Try It Yourself · textbook pages 41–42

04 · Grade Calculator

Write a program that asks for a student's score and prints: "Excellent!" for 90 or above, "Good job!" for 80–89, "Fair effort" for 70–79, "You passed" for 60–69, and "You need to work harder" for below 60. Additionally, if the score is above 100 or below 0, print "Invalid score." Use tryParse() to handle non-numeric input gracefully, and wrap the whole thing in a loop so the user can check multiple scores without restarting.

05 · Vowel Counter

Write a program that takes a word from the user and counts how many vowels it contains (a, e, i, o, and u). Use a for loop to iterate through each character, and use continue to skip consonants. Print the total count at the end. Hint: convert the input to lowercase first with .toLowerCase() to handle both cases.

06 · Enhanced Guessing Game

Modify the guessing game to limit the player to seven attempts. After each wrong guess, tell them how many attempts remain. If they run out, reveal the answer and ask if they'd like to play again. Hint: you can use either a counter with a while loop or replace the while loop with a for loop and break.

Classroom selection: Core tasks are Grade Calculator and Vowel Counter. Enhanced Guessing Game is an extension. These labels are instructor guidance, not part of the textbook exercise wording.